# Isotropic Damage

## Remarks

- If materials are loaded, micro-cracks and/or voids appear, which evolve with increasing load
  ⇝ microscopic damage
- Usually not induced by hydrostatic stress / volume change
- Damage manifests itself phenomenologically by
  - Decrease of material stiffness

"stress-softening" ⇝



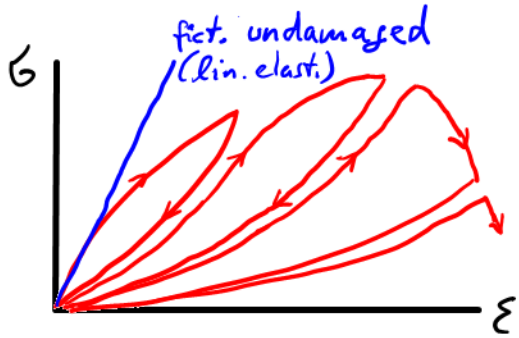

  - Decrease of stress with increase of strain

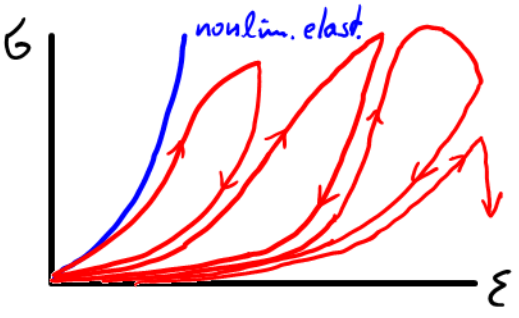
"strain softening" ⇝



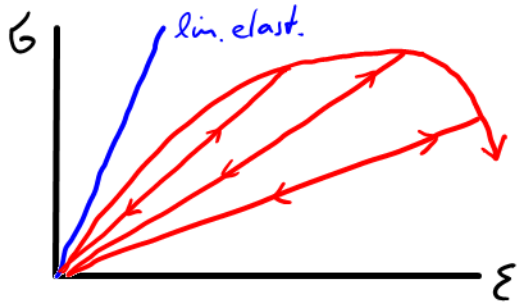

- Pronounced stress-strain hysteresis

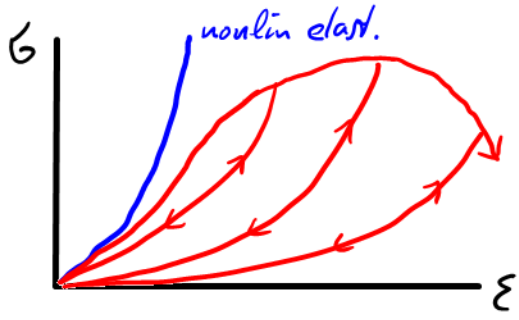
(a) continuous damage



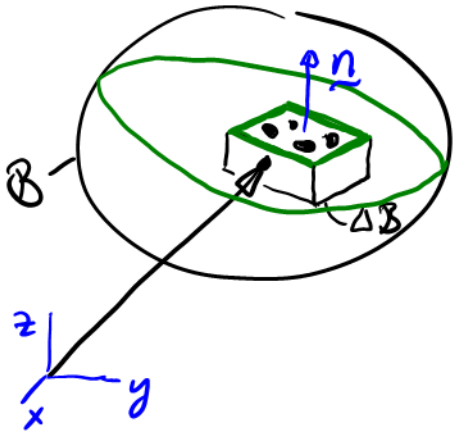



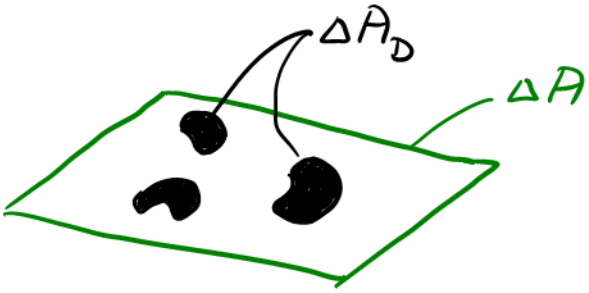

(b) discontinuous damage



# Introduction




- Internal damage state

$$D = D(\underset{\sim}{x}, \underset{\sim}{n}) := \frac{\Delta A_D(\underset{\sim}{x}, \underset{\sim}{n})}{\Delta A(\underset{\sim}{x}, \underset{\sim}{n})} \rightsquigarrow D \in [0, 1]$$

- Concept of equivalent stress

$$N = \sigma \Delta A \overset{!}{=} \sigma^0 (\Delta A - \Delta A_D)$$

$\sigma^0$: stress distributed over $\Delta A_{red} = \Delta A - \Delta A_D$

$\sigma$: stress distributed over $\Delta A$

$N$: resulting normal force in $\Delta A$ in direction $\underset{\sim}{n}$

$$\Leftrightarrow \sigma = \frac{\Delta A - \Delta A_D}{\Delta A} \sigma^0 = \left(1 - \frac{\Delta A_D}{\Delta A}\right) \sigma^0 = (1 - D) \sigma^0$$

$\frac{\Delta A_D}{\Delta A}$: intensity of damage $D$

$\sigma^0$: effective stress (fictitiously undamaged)

# Continuum Damage Mechanics

- Introduction of internal variable for phenomenological description of microstructure change
- Energy

$$\psi := \psi(\underline{\varepsilon}, \alpha) = (1 - D(\alpha))\, \psi^0(\underline{\varepsilon})$$

$\psi^0$: effective energy fct. undamaged

$\alpha$: internal variable

with $D := D(\alpha) \hat{=}$ damage function, $D \in [0, 1]$

$D = 0 \rightsquigarrow$ no damage

$D = 1 \rightsquigarrow$ completely damaged

- Insert time derivative

$$\dot{\psi} = \frac{\partial \psi}{\partial \underline{\varepsilon}} \cdot \dot{\underline{\varepsilon}} + \underbrace{\frac{\partial \psi}{\partial D}}_{= -\psi^0} \frac{\partial D}{\partial \alpha} \dot{\alpha}$$

into Clausius – Duhem – inequality

$$\underline{\sigma} \cdot \dot{\underline{\varepsilon}} - \dot{\psi} \geq 0$$

$$\Leftrightarrow \underline{\sigma} \cdot \dot{\underline{\varepsilon}} - \frac{\partial \psi}{\partial \underline{\varepsilon}} \cdot \dot{\underline{\varepsilon}} + \psi^0 \frac{\partial D}{\partial \alpha} \dot{\alpha} \geq 0$$

$$\Leftrightarrow \left(\underline{\sigma} - \frac{\partial \psi}{\partial \underline{\varepsilon}}\right) \cdot \dot{\underline{\varepsilon}} + \underbrace{\psi^0 \frac{\partial D}{\partial \alpha}}_{\text{"thermodynamic force"}} \dot{\alpha} \geq 0$$

- Standard argument:

$$\rightsquigarrow \underline{\sigma} = \frac{\partial \psi}{\partial \underline{\varepsilon}} = \underbrace{(1 - D(\alpha))\frac{\partial \psi^0}{\partial \underline{\varepsilon}}}_{=: \underline{\sigma}^0}$$

$$\rightsquigarrow \mathcal{D}_{int} = \psi^0 \frac{\partial D}{\partial \alpha}\dot{\alpha} = \psi^0 \dot{D} \geq 0$$

$$\longrightarrow \text{für } \psi^0(\underline{\varepsilon} = \underline{0}) = 0 \rightsquigarrow \psi^0(\underline{\varepsilon}) \geq 0$$

$$\longrightarrow \dot{D} = \frac{\partial D}{\partial \alpha}\dot{\alpha} \geq 0 \quad \text{(no "healing")}$$

- Damage function

  (•) $\psi^0$ is thermodynamic force for $\dot{D}$

  $$\rightsquigarrow D := D(\psi^0) \overset{!}{\hat{=}} D(\alpha) \rightsquigarrow \alpha := \alpha(\psi^0)$$

  (•) $D(\alpha = 0) = 0$

  (•) $D(\alpha = \infty) \in [0, 1]$

- Internal variable:

  (a) continuous damage

  $$\dot{\alpha} = \dot{\alpha}(\psi^0) = |\dot{\psi}^0|$$

  ← at current time $t$

  $$\rightsquigarrow \alpha := \alpha(\psi^0) = \underbrace{\int_t |\dot{\psi}^0| \, dt}_{\text{time integration required}} + \underbrace{\alpha_0}_{=0}$$

(b) Discontinuous damage:

Case distinction

1) Evolution of damage, if

$$\psi^{\circ} \geq \max_{s\,[s<t]} \psi_s^{\circ}, \quad \text{then } \dot{\alpha} = \dot{\psi}^{\circ}$$

2) No evolution of damage, if

$$\psi^{\circ} < \max_{s\,[s<t]} \psi_s^{\circ}, \quad \text{then } \dot{\alpha} = 0$$

↳ damage criterion:

$$\phi := \psi^{\circ} - \alpha \leq 0$$

evolution only if $\phi = 0$

⇳ Internal variable

$$\alpha = \max_{s\in[0,t]} \left[ \psi_s^{\circ} \right]$$

(analytic expression, no time integration of $\dot{\alpha}$ needed)

## Thermodynamic consistence

$$\mathcal{D}_{int} \geq 0 \quad \rightsquigarrow \quad \dot{\mathcal{D}} = \frac{\partial \mathcal{D}}{\partial \alpha}\dot{\alpha} \overset{?}{\geq} 0$$

here: $\dot{\alpha} \geq 0 \Longrightarrow \frac{\partial \mathcal{D}}{\partial \alpha} \overset{!}{\geq} 0$

## Algorithmic treatment

- Stresses: $\underline{\sigma} = (1 - D(\alpha))\frac{\partial \psi^0}{\partial \underline{\varepsilon}}$
- Linearization of stresses $\rightsquigarrow$ tangent moduli

$$\Delta\underline{\sigma} = \frac{\partial \underline{\sigma}}{\partial \underline{\varepsilon}} : \Delta\underline{\varepsilon} + \frac{\partial \underline{\sigma}}{\partial D}\frac{\partial D}{\partial \alpha} \otimes \frac{\partial \alpha}{\partial \underline{\varepsilon}} : \Delta\underline{\varepsilon}$$

$$=: \underline{\mathbb{C}} : \Delta\underline{\varepsilon}$$

with $\underline{\mathbb{C}} = \underbrace{\frac{\partial \underline{\sigma}}{\partial \underline{\varepsilon}}}_{= (1-D(\alpha))\,\underline{\mathbb{C}}^0,\ \underline{\mathbb{C}}^0 = \frac{\partial \underline{\sigma}^0}{\partial \underline{\varepsilon}}} + \underbrace{\frac{\partial \underline{\sigma}}{\partial D}\frac{\partial D}{\partial \alpha} \otimes \frac{\partial \alpha}{\partial \underline{\varepsilon}}}_{\underline{\mathbb{C}}^D}$

$\alpha$ from time integration of evolution equat.

- Introduction of formal time discretization (decomposition of time in time steps)

$$\Delta t = t_{n+1} - t_n$$

For Newton-Raphson, current values at $t_{n+1}$ are needed to be calculated:

$$\left\{ \underline{\sigma}_{n+1},\ \alpha_{n+1},\ \underline{\mathbb{C}}_{n+1} \right\}$$

(a) Continuous damage:

$$\alpha_{n+1} = \int_0^{t_{n+1}} |\dot{\psi}_s^0| \, ds = \underbrace{\int_0^{t_n} |\dot{\psi}_s^0| \, ds}_{\alpha_n} + \underbrace{\int_{t_n}^{t_{n+1}} |\dot{\psi}_s^0| \, ds}_{\Delta\alpha}$$

$$= \alpha_n + \Delta\alpha_{n+1}$$

($\alpha_n$: known from last time step)

$$\text{with } \Delta\alpha_{n+1} = \begin{cases} \left[ \psi_s^0 \right]_{t_n}^{t_{n+1}} & \text{for } \dot{\psi}_s^0 > 0 \\ \left[ -\psi_s^0 \right]_{t_n}^{t_{n+1}} & \text{for } \dot{\psi}_s^0 \leq 0 \end{cases}$$

$$= \begin{cases} \left( \psi_{n+1}^0 - \psi_n^0 \right) & \text{for } \dfrac{\psi_{n+1}^0 - \psi_n^0}{\Delta t} > 0 \quad (\geq 0) \\ -\left( \psi_{n+1}^0 - \psi_n^0 \right) & \text{for } \dfrac{\psi_{n+1}^0 - \psi_n^0}{\Delta t} \leq 0 \quad (\geq 0) \end{cases}$$

$$= \left| \psi_{n+1}^0 - \psi_n^0 \right|$$

($\psi_n^0$: known from last time step)

$$\rightsquigarrow \quad D_{n+1} = D\big|_{\alpha_{n+1}}$$

$$\rightsquigarrow \quad \underline{\sigma}_{n+1} = (1 - D_{n+1}) \, \underline{\sigma}^0 \big|_{\underline{\varepsilon}_{n+1}}$$

$$\rightsquigarrow \quad \underline{\mathbb{C}}_{n+1} = \left[ (1-D)\underline{\mathbb{C}}^0 + \frac{\partial D}{\partial \alpha} \frac{\partial \sigma}{\partial D} \otimes \frac{\partial \alpha}{\partial \underline{\varepsilon}} \right]_{\underline{\varepsilon}_{n+1}}$$

with $\frac{\partial \underline{\sigma}}{\partial D} = \frac{\partial (1-D)\underline{\sigma}^0}{\partial D} = -\underline{\sigma}^0$

$$\frac{\partial \alpha}{\partial \underline{\varepsilon}} = \begin{cases} \frac{\partial \psi^0}{\partial \underline{\varepsilon}} = \underline{\sigma}^0 & \text{if} \quad \psi^0_{n+1} - \psi^0_n \geq 0 \\ -\frac{\partial \psi^0}{\partial \underline{\varepsilon}} = -\underline{\sigma}^0 & \text{if} \quad \psi^0_{n+1} - \psi^0_n < 0 \end{cases}$$

$$\Rightarrow \underline{\mathbb{C}}_{n+1} = \begin{cases} (1-D)\underline{\mathbb{C}}^0 - \frac{\partial D}{\partial \alpha} \underline{\sigma}_0 \otimes \underline{\sigma}_0 & \text{if} \quad \psi^0_{n+1} - \psi^0_n \geq 0 \\ (1-D)\underline{\mathbb{C}}^0 + \frac{\partial D}{\partial \alpha} \underline{\sigma}_0 \otimes \underline{\sigma}_0 & \text{if} \quad \psi^0_{n+1} - \psi^0_n < 0 \end{cases}$$

(b) Discontinuous damage

(i) no evolution if $\phi_{trial} := \psi^0_{n+1} - \alpha_n \leq 0$

(new $\psi^0 \leq$ current maximum)

$\rightarrow \alpha_{n+1} = \alpha_n$

$\rightarrow \underline{\mathbb{C}} = (1-D)\underline{\mathbb{C}}^0$ (since $\dot{D} = 0 \rightsquigarrow \underline{\mathbb{C}}^D = 0$)

(ii) damage evolution if $\phi_{trial} > 0$

$\rightarrow \alpha_{n+1} = \psi^0_{n+1}$

$\rightarrow \underline{\mathbb{C}} = (1-D)\underline{\mathbb{C}}^0 + \underline{\mathbb{C}}^D$

with $\underline{\mathbb{C}}^D = \frac{\partial \underline{\sigma}}{\partial D} \frac{\partial D}{\partial \alpha} \otimes \frac{\partial \alpha}{\partial \underline{\varepsilon}}$

$= -\frac{\partial D}{\partial \alpha} \underline{\sigma}_0 \otimes \underline{\sigma}_0$

# Example 1: Isochoric continuous damage

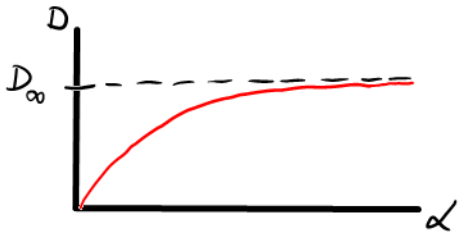
- $\psi = \frac{1}{2}\varkappa\,(\mathrm{tr}\,\underline{\varepsilon})^2 + (1-D)\,\psi^0$

  with $\psi^0 = \mu \|\mathrm{dev}\,\underline{\varepsilon}\|^2$ (isochoric damage)

- $\alpha_{n+1} = \alpha_n + \left|\psi^0_{n+1} - \psi^0_n\right|$

- $D = D_\infty \left[1 - \exp\left(-\frac{\alpha}{D_{s0}}\right)\right]$

  damage parameters




$D_\infty \in [0, 1]$

$D_{s0} > 0$

- $\frac{\partial D}{\partial \alpha} = \frac{D_\infty}{D_{s0}} \exp\left(-\frac{\alpha}{D_{s0}}\right) \geq 0$ ✓ thermodynamically consistent

- $\underline{\sigma} = \varkappa\,(\mathrm{tr}\,\underline{\varepsilon})\,\underline{1} + 2(1-D)\,\mu\,\mathrm{dev}\,\underline{\varepsilon}$

- $\underline{\mathbb{C}} = \varkappa\,\underline{1}\otimes\underline{1} + 2(1-D)\,\mu\,\underline{\mathbb{P}} + \underline{\mathbb{C}}_D$

with $\underline{\mathbb{C}}_D = \begin{cases} -4\mu^2 \frac{\partial D}{\partial \alpha}\,\mathrm{dev}\,\underline{\varepsilon}\otimes\mathrm{dev}\,\underline{\varepsilon} & \text{if } \psi^0_{n+1} - \psi^0_n \geq 0 \\ 4\mu^2 \frac{\partial D}{\partial \alpha}\,\mathrm{dev}\,\underline{\varepsilon}\otimes\mathrm{dev}\,\underline{\varepsilon} & \text{if } \psi^0_{n+1} - \psi^0_n < 0 \end{cases}$

history variables $\alpha$ and $\psi^0$ must be stored for next time step!

# Example 2: Isochoric discontinuous damage

- $\psi$, $\psi_0$ (see expl. 1)
- $\psi^0 = \mu \| \operatorname{dev} \underline{\varepsilon}_{n+1} \|$
- $\phi_{trial} = \psi^0 - \alpha_n$

  (i) $\phi_{trial} \leq 0 \rightsquigarrow \alpha_{n+1} = \alpha_n$

  (ii) $\phi_{trial} > 0 \rightsquigarrow \alpha_{n+1} = \mu \| \operatorname{dev} \underline{\varepsilon}_{n+1} \|^2$
- $D = D_\infty \left[ 1 - \exp\left( -\frac{\alpha}{D_{s0}} \right) \right]$
- $\frac{\partial D}{\partial \alpha} = \frac{D_\infty}{D_{s0}} \exp\left( -\frac{\alpha}{D_{s0}} \right)$
- $\underline{\sigma} = \lambda (\operatorname{tr} \underline{\varepsilon}) \underline{1} + 2(1-D)\mu \operatorname{dev} \underline{\varepsilon}$
- (i) $\underline{\mathbb{C}} = \lambda \underline{1} \otimes \underline{1} + 2(1-D)\mu \underline{\mathbb{P}}$

  (ii) $\underline{\mathbb{C}} \Leftarrow \underline{\mathbb{C}} + \underline{\mathbb{C}}^D$

  with $\underline{\mathbb{C}}^D = -4\mu^2 \frac{\partial D}{\partial \alpha} \operatorname{dev} \underline{\varepsilon} \otimes \operatorname{dev} \underline{\varepsilon}$

history variable $\alpha$